

Machine Tool Technology

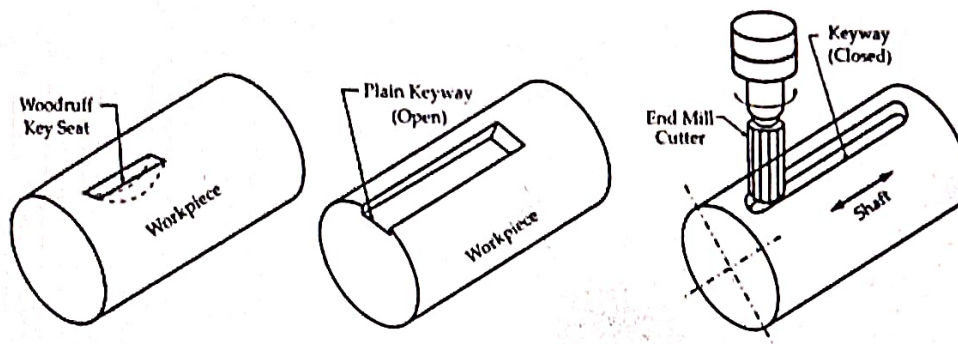
Week-10

1. Explain Key way and V- slot Operation
2. Prepare the Process plan as per the given drawing for Key way and V-slot
3. Estimate the Production cost per Piece as per the given drawing for Key way and V-slot

Key way and V- slot Operation

On a milling machine, keyway milling creates slots (keyways) for keys that secure rotating parts. The key and keyway play a significant role when it comes to a gear or similar rotating equipment needing to be fitted onto a rod or shaft in order to transfer rotational force. While V-slot milling creates V-shaped grooves, often used for clamping or holding workpieces.

Keyway Milling:



Purpose:

Keyways are slots cut into shafts to accommodate keys, which prevent relative rotation between the shaft and a connected component (like a gear or pulley).

Tools:

Keyway milling uses specialized cutters, like end mills, side milling cutters, or metal slitting saws, depending on the keyway shape and size.

Process:

Roughing: A key slot milling operation roughs out the full slot.

Finishing: Side milling is used to finish the slot, creating true square corners.

Accuracy: For best accuracy, the operation may involve using an undersized end mill and dividing the process into two steps: roughing and finishing.

Example:

Woodruff key slotting, where a Woodruff slot cutter creates the keyway or key seat for a Woodruff key.

Clamping:

To ensure workpiece axis groove symmetry, clamping methods include:

Flat clamp

V-shaped block

Dividing head clip

V-Slot Milling:

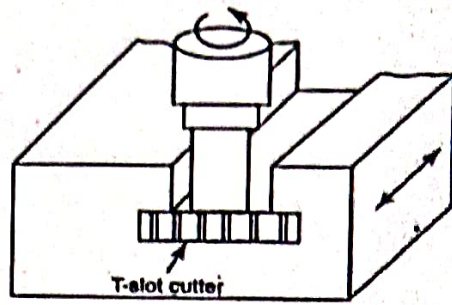


Figure 3.131 T-slot milling

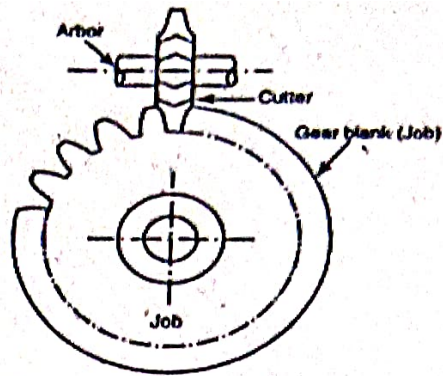


Figure 3.132 Gear milling

Purpose:

V-slots, or V-grooves, are used for various applications, including:

Clamping workpieces

Holding workpieces during machining

Creating a guide for a moving part

Tools:

V-slot milling utilizes cutters with a V-shaped profile, such as V-groove cutters or angle milling cutters.

Process:

The cutter is positioned to match the desired V-slot angle.

The workpiece is clamped securely.

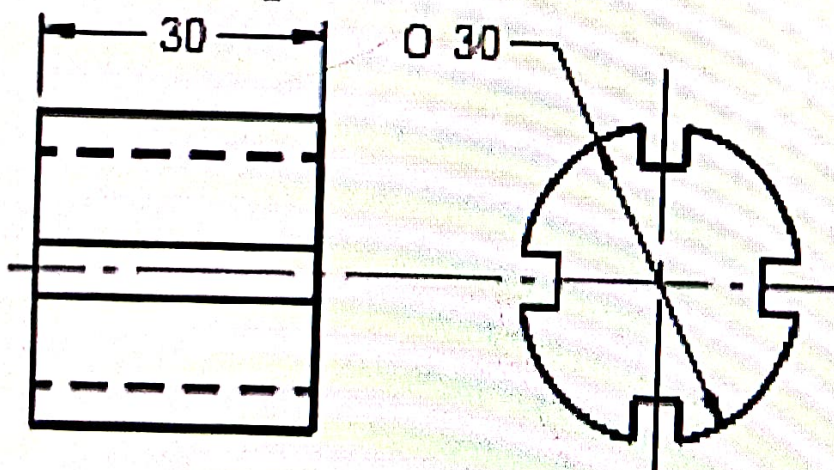
The milling machine table is moved to cut the V-slot.

Example:

Creating a V-groove on a workpiece for clamping or as a guide for a sliding component.

Q-Mill Keyways on cylinder surface of $\phi 30$ mm for length 30 mm as per given sketch, with process plan and estimate the cost of production. The width and depth of the keyway is 5 mm and 4 mm respectively. Raw Material Cost = Rs. 11/ piece. Density of raw material = 7.9gm./cc. Cost of Scrap = Rs. 20/kg. Labour Cost = Rs. 100/hour. For milling, Feed = 30 mm/min., Depth of cut = 1 mm/ per cut. Overall setup and indexing time = 15 mins. Machining Cost = Rs. 200/hr. Overhead charges = 25% of Material Cost. Consider suitable missing data.

All Dimensions are in mm



Solution

Step by Step Solution:

Step 1

Calculate the volume of the keyway to be removed:

$$\text{Volume} = \text{width} \times \text{depth} \times \text{length} = 5 \text{ mm} \times 4 \text{ mm} \times 30 \text{ mm} = 600 \text{ mm}^3.$$

Step 2

Convert the volume to cm^3 :

$$\text{Volume} = 600 \text{ mm}^3 \times (1 \text{ cm}^3 / 1000 \text{ mm}^3) = 0.6 \text{ cm}^3.$$

Step 3

Calculate the mass of the material to be removed:

$$\text{Mass} = \text{Volume} \times \text{Density} = 0.6 \text{ cm}^3 \times 7.9 \text{ g/cm}^3 = 4.74 \text{ g}.$$

Step 4

Convert the mass to kg: $\text{Mass} = 4.74 \text{ g} \times (1 \text{ kg} / 1000 \text{ g}) = 0.00474 \text{ kg}.$

Step 5

Calculate the cost of scrap:

$$\text{Scrap Cost} = \text{Mass} \times \text{Cost per kg} = 0.00474 \text{ kg} \times \text{Rs. } 20/\text{kg} = \text{Rs. } 0.0948.$$

Step 6

Calculate the machining time:

$$\text{Number of passes} = \text{Depth of keyway} / \text{Depth of cut per pass} = 4 \text{ mm} / 1 \text{ mm} = 4 \text{ passes}.$$

$$\text{Machining time per pass} = \text{Length} / \text{Feed rate} = 30 \text{ mm} / 30 \text{ mm/min} = 1 \text{ min}.$$

$$\text{Total machining time} = 4 \text{ passes} \times 1 \text{ min/pass} = 4 \text{ mins}.$$

Step 7

Calculate the total labour cost:

$$\text{Labour Cost} = \text{Labour rate} \times \text{Total time} = \text{Rs. } 100/\text{hour} \times (4 \text{ mins} / 60 \text{ mins}) = \text{Rs. } 6.67.$$

Step 8

$$\begin{aligned} \text{Calculate the total machining cost: } \text{Machining Cost} &= \text{Machining rate} \times \text{Total time} = \text{Rs. } 200/\text{hour} \\ &\times (4 \text{ mins} / 60 \text{ mins}) = \text{Rs. } 13.33. \end{aligned}$$

Step 9

$$\begin{aligned} \text{Calculate the overhead charges: } \text{Overhead Charges} &= 25\% \text{ of Material Cost} = 0.25 \times \text{Rs. } 11 = \\ &\text{Rs. } 2.75. \end{aligned}$$

Step 10

Calculate the total cost of production:

$$\begin{aligned} \text{Total Cost} &= \text{Raw Material Cost} + \text{Scrap Cost} + \text{Labour Cost} + \text{Machining Cost} + \text{Overhead} \\ \text{Charges} &= \text{Rs. } 11 + \text{Rs. } 0.0948 + \text{Rs. } 6.67 + \text{Rs. } 13.33 + \text{Rs. } 2.75 = \text{Rs. } 33.8448. \end{aligned}$$

Final Answer:

The total cost of production for milling the keyways on the cylinder surface is Rs. 33.8448.